

Assignment 1- Part 2 Logical Design

This assignment will outline and complete the steps for normalization of the information provided in the assignment spec.

Note: You can scroll through the doc form start to finish to follow the process logically from ONF to 3NF. To view the actual tables in each form, look at the table immediately under the section “**Entities and Attributes- (X)NF**”. You can use the navigation headers to click directly to these locations if you need.

Provided detail

Below is the provided detail, a student details form.

Student Details:				Date: 10/12/2023
Student No:	99123456			
Student Name:	Joe Bloggs			
Course Enrolled:	CBEM			
Mode of Study:	Full Time			
Student Advisor No:	A124			
Student Advisor Name:	A. N. O'Reilly			
Subject Details:				
Subject Number	Subject Name	Year/Semester	Grade	
COMP 335	Databases	2022/1	A	
COMP 304	Operating Systems	2022/2	B	

Step 1: Place in UNF.

To place in UNF we have to do the following:

- 1) Identify the name of the entity.
- 2) Identify all attributes
- 3) Identify repeating groups
- 4) Select the primary Key
- 5) Identify if there are any calculable values, those which don't need to be recorded as attributes in the database

Students will be the most appropriate name of the Entity, and the values from the student record form will translate into attributes.

From the data presented, some tables also contain multiple entries or repeating groups. These are highlighted in the colour **green**.

Entity and Attributes – UNF

The following data is in UNF:

Students (**Stud No.**, Stud name, course enrolled, Mode of study, Student Advisor No., Stud Advisor name, **Subj Num**, **Subj name**, **Year/semester**, **Grade**)

The following colour key is used throughout

Blue= Entity

Bold and underlined = PK

Green = Repeating groups

Process Reasoning:

- 1) The attributes **Subj Num**, **Subj name**, **Year/semester**, **Grade** attributes can all contain multiple values, as such they are deemed repeating groups.
- 2) "Stud No." will be a unique identifier, as such that will be appropriate to use as a Primary Key (PK) to uniquely identify data.
- 3) There are no calculable values.

Step 2: Place in 1NF.

To place in 1NF we have to do the following:

- 1) Remove repeating Groups
- 2) Identify an appropriate PK to use in any removed repeating group.

Entity and Attributes – 1NF

The following data is in 1NF:

Students-1(**Stud No.**, Stud name, course enrolled, Mode of study, Student Advisor No., Stud Advisor name)

Student-subject-data-1((**Stud No.**, **Subj Num**) **Subj name**, **Year/semester**, **Grade**)

Process Reasoning:

- 1) Repeating groups have been removed from **Students-1** and placed in their own table **Student-subject-data-1**
- 2) "Subject Num" is a unique value and as such this is suitable to identify attributes in the new table. However, we need a unique identifier to identify the grade with the student and maintain that table relationship. As such we will move the "Stud No". over to the **Student-subject-data-1** table to make a composite Primary Key (PK) of "Stud No", + "Subj No".
- 3) We also append '**-1**' to the table name to keep indicate they are in 1NF.

Step 3: Place in 2NF

To place in 2NF we have to do the following:

- 1) Identify and remove Partial dependencies i.e. every non key attribute has to be fully dependent on the primary key
- 2) We do this for each table in turn.

Partial dependencies

Table 1 – Partial dependencies

Students-1(Stud No., Stud name, course enrolled, Mode of study, Student Advisor No., Stud Advisor name)

- There are no partial dependencies in Table 1
 - a. No composite key exists and as such all attributes are fully dependent on the PK, meaning there are no partial dependencies.

Table 2 – Partial dependencies

Student-subject-data-1(Stud No., Subj Num), Subj name, Year/semester, Grade)

- There is a partial dependency in table 2
 - a. “Subj name” is partially dependent on the PK, it is dependent on the “Subj num”.
 - b. We will create a new table called **Subjects** into which we will place the partially dependent attributes, and rename the existing table **Grades**. This is to differentiate the table names and keep them relevant to the data they contain.
 - c. We will also rename the table **Student-subject-data** to **Subjects** to keep the table name relevant and easily identifiable to the data it obtains.

Entities and Attributes – 2NF

The following data is in 2NF:

Students-2(Stud No., Stud name, course enrolled, Mode of study, Student Advisor No., Stud Advisor name)

Grades-2(Stud No., Year/semester, Grade)

Subjects-2(Subj Num, Subj name)

Process Reasoning:

- We remove the partial dependency, identify appropriately unique PKs and renamed the tables to keep the relevant and identifiable.
- We also append ‘-2’, to the table name to keep indicate they are in 2NF

Step 4: Place in 3NF

To place in 3NF we have to do the following:

- 1) Identify and remove **Transitive dependencies** i.e. every non key attribute has to be fully dependent on the primary key, not also having a dependency on another attribute.

2) We do this for each table in turn.

Transitive dependencies

Table 1

- **Student-2**(Stud No., Stud name, course enrolled, Mode of study, Student Advisor No., Stud Advisor name)
- There are two transitive dependencies present in table 1
 - 1) “**Mode of study**” has a transitive dependency on “**course enrolled**”. Not all courses will offer the same mode of study. It will be course dependent.
 - i. We will create a new unique identifier to act as PK for this new table, “**Course No.**”.
 - ii. To address this transitive dependency, we will create a new table called **Courses** and place those related attributes with transitive dependencies with it.
 - 2) “**Stud Advisor name**” has a transitive dependency on “**Student Advisor No.**”
 - i. To address this separate transitive dependency, we will create a new table called **Advisors** and place those attributes with transitive dependencies with it.
 - ii. We use the “**Student Advisor No.**” as the PK in the new table **Advisors**. This is a unique value which will allow us uniquely identify the student advisor.

Table 2

Grades-2(Stud No., Year/semester, Grade)

- There are no transitive dependencies in Table 2.

Table 3

Subjects-2(Subj Num, Subj name)

- There are no transitive dependencies in Table 3.

Entities and Attributes – 3NF

The following data is in 3NF:

Students-3(Stud No., Stud name)

Courses-3(Course No., course enrolled, Mode of study)

Advisors-3(Student Advisor No., Student Advisor name)

Grades-3(Stud No., Year/semester, Grade)

Subjects-3(Subj Num, Subj name)

Normalisation Assumptions:

- We assume a student can only enrol in one course semester/year.
- Students can enroll in multiple subjects per semester year.
- Students can only have one grade per subject per year. There is no grade per course.
- We assume advisors can be allocated many students to act as advisor to but students will be allocated only one advisor.
- "Course enrolled" attribute does not appear to be a reliable unique identifier, due to possible duplication at some point. As such will create a new unique identifier for **Course-data-3**, call "Course No.". This will allow us uniquely identify each course.
- Grade has two unique identifiers in student and course, hence requires a composite PK.
- Mode of study will have a dependency on course as enrolled as not all courses will offer full time or part time study.
- Although not called out in initial spec. some tables look like they do not have a lot of data however we will assume additional data will be present. For example:
 - **Student** attributes would also include be name, address, email, phone number etc
 - **Advisors** data would include address, contact details etc
- **Advisors** may or may not be the same as lecturers, we will assume advisors are not necessarily the same as lecturers, as such will name them as 'advisors', not 'lecturers'.
- Assume grades will be stored as an alphabet value, A, B, C etc not a number or percentage, descriptive value.
- We have not included Foreign Keys (FKs) in the tables however they would need to be included to maintain the relationships between the tables for example:
 - "Subj num" would be a FK in the **Grades** table, to maintain an efficient relationship between student, grade and subject data.
 - "Student Advisor Num" would be a FK in the **Student** table

Improvements

To improve we could consider the following:

- Clarity around the advisor/lecture relationship
- Clarity around the grading format and data values for each course type, arts, science, business etc and any possible differences.
- Account for students leaving one course and enrolling in another within the same semester
- Repeat exams and having different grades in the same academic year. Perhaps the terms are defined by 3 semesters a year and grades associated with each of those per subject.
- Growth of data, scale of students and course data to be stored.
- Most common queries identified to allow for appropriate relationships between tables to be established for efficient return of data.